Sepehr Samavi

Ph.D. Candidate, University of Toronto

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(Last updated June 18, 2026)

Education

Ph.D., University of Toronto *since 01/2021*
Institute for Aerospace Studies
Advisors: Angela P. Schoellig, Florian Shkurti
Committee: Timothy D. Barfoot, Jonathan S. Kelly

M.A.Sc., University of Toronto *11/2021*
Institute for Aerospace Studies, Emphasis in Robotics
Title: Accounting for Unpredictability in Autonomous Driving Behaviour [pdf]
Advisor: Angela P. Schoellig

B.A.Sc. in Engineering Science (with Honours), University of Toronto *06/2018*
Division of Engineering Science, Robotics Engineering Major
Thesis Title: Lane Detection for an Autonomous Vehicle [pdf]
Thesis Advisor: Angela P. Schoellig

Affiliations and Leadership Roles

Graduate Student Researcher, Vector Institute for Artificial Intelligence *since 01/2020*
An independent, not-for-profit corporation that is part of the Pan-Canadian AI Strategy dedicated to advancing the growth of Artificial Intelligence in Canada. Researchers have access to computing, office space, and a research grant among other resources. Member of Space Planning Committee (Summer 2023). [site]

Graduate Student Researcher, University of Toronto Robotics Institute *since 06/2019*
An interdepartmental research and teaching institute for fostering collaboration among researchers in robotics and related areas. [site]

Visiting Researcher, Technical University of Munich *11/2024 - 10/2025*
At the Chair of Safety, Performance and Reliability for Learning Systems. [site]

Deputy Lead, aUToronto, U of T's self-driving car team *09/2018 - 09/2019*
Competing in the General Motors/SAE AutoDrive Challenge, a three-year competition to develop a self-driving car (2017-2020). The team consisted of around 80 undergraduate students and a few graduate students advised by faculty. As a graduate student I helped the team-lead with preparing for the second year of the competition. Previously as an undergraduate, I was an autonomy software developer for the first year of the competition (09/2017 - 06/2018). [site]

Awards and Honours

Ontario Graduate Scholarship, $15K *09/2024 - 09/2025*
Government of Ontario, *Scholarship for one year of Ph.D. research.*

Alexander Graham Bell Canada Graduate Scholarship, $105K *01/2021 - 12/2023*
Natural Sciences and Engineering Research Council, Government of Canada
Scholarship for three years of Ph.D. research.

Ontario Graduate Scholarship, $10K *09/2019 - 04/2020*
Government of Ontario, *Scholarship for M.A.Sc. research.*

Vector Scholarship in Artificial Intelligence, $17.5K *09/2018 - 09/2019*
Vector Institute, *Scholarship for M.A.Sc. research.*

Exceptional Opportunities Award, $3K *06/2014 - 08/2014*
University of Toronto, *Funding for undergraduate research at Harvard Medical School.*

President's Entrance Scholarship, $2K *09/2013*
University of Toronto, *Entrance scholarship for undergraduate studies.*

Best Poster Presentation *06/2019*
International Conference on Robots and Vision (CRV)
as the presenting author for publication [C3].

First Place in 2019 SAE AutoDrive Challenge *06/2019*
GM and SAE International, Second year of three-year competition
as a deputy lead of the University of Toronto team, aUToronto.

First Place in 2018 SAE AutoDrive Challenge *06/2018*
GM and SAE International, First year of three-year competition
as a member of the University of Toronto team, aUToronto.

Publications

An up-to-date list of my publications is available on my Google Scholar profile.

Journal Impact Factors (JIF) scores were obtained from Web of Science Journal Info 06/2025.

Peer Reviewed - Journal Publications

[J1] **S. Samavi**, A. Lem, F. Sato, S. Chen, Q. Gu, K. Yano, A. P. Schoellig, and F. Shkurti, "SICNav-Diffusion: Safe and Interactive Crowd Navigation with Diffusion Trajectory Predictions," *IEEE Robotics and Automation Letters (RA-L)*, vol. 10, no. 9, pp. 8738–8745, 2025. [JIF 5.3] [site][code][arXiv]

[J2] **S. Samavi**, J. R. Han, F. Shkurti, and A. P. Schoellig, "SICNav: Safe and Interactive Crowd Navigation using Model Predictive Control and Bilevel Optimization," *IEEE Transactions on Robotics (T-RO)*, vol. 41, pp. 801–818, 2024. [JIF 10.5] [site][code][arXiv]

[J3] K. Burnett, J. Qian, X. Du, L. Liu, D. J. Yoon, T. Shen, S. Sun, **S. Samavi**, M. J. Sorocky, M. Bianchi, K. Zhang, A. Arkhangorodsky, Q. Sykora, S. Lu, Y. Huang, A. P. Schoellig, and T. D. Barfoot, "Zeus: A system description of the two-time winner of the collegiate SAE AutoDrive competition," *Journal of Field Robotics*, vol. 38, no. 1, pp. 139–166, 2021. [url][pdf]

Peer Reviewed - Conference Proceedings

[C1] F. D'Orazio*, **S. Samavi***, X. Du*, S. Zhou, G. Oriolo, and A. P. Schoellig, "SM^{2}ITH: Safe Mobile Manipulation with Interactive Human Prediction via Task-Hierarchical Bilevel Model Predictive Control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **equal contribution.* [site][code][arXiv]

[C2] **S. Samavi**, F. Shkurti, and A. P. Schoellig, "Does unpredictability influence driving behavior?" in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 1720–1727. [url][pdf]

[C3] K. Burnett, **S. Samavi**, S. L. Waslander, T. D. Barfoot, and A. P. Schoellig, "aUToTrack: A Lightweight Object Detection and Tracking System for the SAE AutoDrive Challenge," in *Conference on Computer and Robot Vision (CRV)*, 2019, pp. 209–216. **Best Poster Presentation Award.** [url][pdf]

[C4] K. Burnett, A. Schimpe, **S. Samavi**, M. Gridseth, C. W. Liu, Q. Li, Z. Kroeze, and A. P. Schoellig, "Building a Winning Self-Driving Car in Six Months," in *IEEE International Conference on Robotics and Automation (ICRA)*, May 2019, pp. 9583–9589. [url][pdf]

Refereed Abstracts and Presentations

[A1] N. Schlüter, M. Schuck, L. Brunke, **S. Samavi**, S. Zhou, and A. P. Schoellig, "MAD-IRL: Multi-Agent Drone Racing Using Inverse RL and Iterative Best-Response MPC," in *German Robotics Conference (GRC)*, 2026. [pdf]

[A2] **S. Samavi**, J. R. Han, F. Shkurti, and A. P. Schoellig, "SICNav: Safe and Interactive Crowd Navigation using Model Predictive Control and Bilevel Optimization," in *AAAI Conference on Artificial Intelligence (AAAI)*, 2026, abstract reprint and presentation of the IEEE T-RO publication. [abstract][presentation][arXiv]

[A3] **S. Samavi**, G. Bhutani, F. Shkurti, and A. P. Schoellig, "Deploying SICNav in the Field: Safe and Interactive Crowd Navigation using MPC and Bilevel Optimization," in *Workshop on Field Robotics at the IEEE International Conference on Robotics and Automation (ICRA W-FR)*, 2025. [arXiv]

[A4] **S. Samavi**, A. Lem, F. Sato, S. Chen, Q. Gu, K. Yano, A. P. Schoellig, and F. Shkurti, "Safe Diffusion Model Predictive Control for Interactive Robotic Crowd Navigation," in *Workshop on Safe and Robust Robot Learning for Operation in the Field at the Conference on Robot Learning (CoRL)*, 2024. [pdf]

Employment History

Technical University of Munich, Munich, Germany *11/2024 - 10/2025*
Chair of Safety, Performance and Reliability for Learning Systems,
Visiting Research Fellow (Wissenschaftlicher Mitarbeiter)

University of Toronto, Toronto, Canada *since 01/2021*
Division of Engineering Science
Teaching Assistant, *Artificial Intelligence ROB311*

Magna International, Greater Toronto Area, Canada *05/2016 - 07/2017*
Magna Electronics Vision Centre
Algorithms Research Intern, *Perception for Magna Automated Parking Project*

Harvard Medical School, Boston, USA *06/2014 - 08/2014*
Tearney Lab, Wellman Center for Photomedicine
Undergraduate Research Intern, *Prototyping for gastrointestinal tethered capsule retraction device*

Advising and Mentorship

Masters Thesis Supervision
Niklas Schlüter, Technical University of Munich (with Lukas Brunke and Martin Schuck) *2024-2025*
Opponent Modeling for Multi-Drone Racing via Inverse Reinforcement Learning and Model Predictive Control

Undergraduate Thesis Supervision
James R. Han, Engineering Science, University of Toronto (with Prof. Florian Shkurti) *2022-2023*
Investigating Reinforcement Learning Based Control for Sidewalk Navigation
Nikhil Thiyagarajan, Engineering Science, University of Toronto (with Prof. Florian Shkurti) *2025-2026*
Multi-Robot Safe and Interactive Crowd Navigation

Undergraduate Summer Student Supervision
Xinyuan (Sam) Qiao, Engineering Science, University of Toronto *Summer 2021*
Ruichen (Spencer) Li, Elec. and Comp. Engineering, University of Toronto *Summer 2023*
Garvish Bhutani, Engineering Science, University of Toronto *Summer 2024*

Invited Talks

MARMoT Lab, National University of Singapore (NUS) *01/2026*
Invited talk at internal research seminar (Prof. Guillaume Sartoretti), "Safe and Interactive Crowd Navigation"

ATARI Lab, Technical University of Munich (TUM) *08/2025*
Invited talk at internal research seminar (Prof. Majid Khadiv), "Safe and Interactive Crowd Navigation"

University of Toronto Robotics Institute 2021 Autonomous Vehicles Workshop *06/2021*
Student talk, "Accounting for Unpredictability in Autonomous Driving Behaviour" [video]

Professional Activities and Service

Workshop Organization
Co-organizer of *Workshop on Semantics for Robotics: From Environment Understanding and Reasoning to Safe Interaction*, Robotics: Science and Systems (RSS) [event page] *2024*

Committee Service
Student Experience Committee, Univ. of Toronto Inst. for Aerospace Studies *2023-2024*
Space Planning Committee, Vector Institute for Artificial Intelligence *2023*

Reviewing

IEEE International Conference on Robotics and Automation (ICRA)	*since 2020*
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	*since 2020*
IEEE Robotics and Automation Letters (RA-L)	*since 2022*
Conference on Robot Learning (CoRL)	*since 2023*

Media Coverage

ZDF moma future: Robot learns how to walk like a human (in German). The German public broadcaster, ZDF, produced a news story about our crowd navigating robot aired on their morning news program, Morgenmagazin. Original title: "moma future: Roboter lernt Menschenlauf." By Moritz Neuß. [published video] [with English subtitles] *05/2025*

heise online: Robot moves safely through crowds of people. heise online, a leading German technology news outlet, reported on our crowd-navigating robot. German title: "Roboter bewegt sich sicher durch Menschenmengen." By Oliver Bünte. [article][German version] *03/2025*

TUM: Robot Jack moves like a human. The Technical University of Munich (TUM) published a press release about our crowd-navigating robot, "Jack." [press release] [German version] *03/2025*

IEEE Spectrum Video Friday: Robot Holiday Videos. A short Christmas-themed video of our crowd navigation robot was featured in IEEE Spectrum's annual holiday robot-video roundup. By Evan Ackerman. [video][article] *12/2024*

AI: Hype vs. Reality. S1E1 AI That Drives: Self-Driving Cars. A popular science podcast and video by Dell Technologies about emerging AI technologies recorded an episode about our self-driving car team competing in the AutoDrive challenge. By Jessica Chobot. [audio] *09/2019*

Outreach and Volunteering

Civic Engagement *12/2021*

I helped draft a letter that was submitted to the Toronto City Council on behalf of the University of Toronto Robotics Institute to advocate for policies that enable the responsible development of service robots in the city. [press release][pdf]

STEM outreach & live demonstrations *09/2018 - present*

I often volunteer to show live demonstrations of our robots to visitors, who have included government officials, representatives from industry, journalists, professors and their research groups, elementary and high-school students (e.g. 40 students aged 10-12 years and 30 high school students in 11/2019), as well as undergraduate and graduate students.

Skills and Languages

Programming and Scripting: Python, C/C++, Java, MATLAB, bash

Scientific Computing: Un*x (macOS, Linux), ROS, Docker, Git, Slurm

Machine Learning & Optimization: PyTorch, JAX, CasADi, Acados

Languages: English (native), Persian (native), French (fluent)